

YASH BAVADIYA

+91-9879163975 krbavadiya11@gmail.com [Portfolio](#) [LinkedIn](#) [GitHub](#) [LeetCode](#)

EDUCATION

Indian Institute of Technology, Jodhpur

Aug 2024 – Jul 2028

Bachelor's Degree - Computer Science and Engineering - CGPA - 7.21

Jodhpur, Rajasthan

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, MySQL

Technologies/Frameworks: React, Next.js, Express, Prisma, Qt (C++), PostgreSQL, MongoDB, Redis, WebSockets, WebRTC, Firebase, Docker, Linux/UNIX environments, Relational Databases, REST APIs, TCP/IP

Tools: Git, Nginx, GitHub Actions, Vercel, Postman, VS Code

Core Competencies: Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, System Design, Distributed Systems, REST APIs

EXPERIENCE

Google Summer of Code 2026

May 2026 – Jun 2026

Kdenlive @KDE Community | Open Source Contributor, C++ and Qt

Remote

- Secured selection for **GSoC 2026** among 1,200 global contributors; implementing Kdenlive's **RGB curve editor, gradient editor, and speed ramp widgets** in C++ and Qt under lead maintainer Jean-Baptiste Mardelle.
- Integrated **20 contributions** across Kdenlive, KDE Documentation, and Media Lovin' Toolkit upstream; resolved an **11-year-old bug** via a union-of-intervals gap-detection algorithm shipping to **41,000 monthly Flathub installs**.

Undergraduate Researcher

Jul 2025 – Nov 2025

Indian Institute of Technology, Jodhpur | Supervisor: Dr. Pratik Mazumder

IIT Jodhpur

- Engineered an offline **Retrieval-Augmented Generation pipeline** integrating FAISS vector search with 4-bit quantized Phi-3 and Flan-T5-XL; fine-tuned model **outperformed base on all 8 ROUGE metrics**, achieving **+71.77% average improvement**.
- Attained **3 second query latency** on 300 page PDFs; deployed to **15 IIT Jodhpur students** during semester exams across 3 feedback-driven chunking iterations.

PROJECTS

Peek-a-Repo | [GitHub](#) | [Web Store](#) | TypeScript, GitHub GraphQL API, Webpack

Feb 2026 – May 2026

- Deployed a Chrome extension on the Web Store rendering GitHub file contents inline on hover via GraphQL API and **Document Object Model injection; 88% monthly growth** in store views across US, Japan, and India.
- Programmed SHA-keyed cache with debounced fetch maintaining GitHub's **5,000 requests per hour** rate limit; **5 weekly active users** across Windows, Linux, and macOS with zero reported bugs.
- Optimized via Webpack bundling, reducing package size by **60%**; enforced content security policy headers and sandboxed execution to pass Chrome Web Store security review.

DaemonDoc | [GitHub](#) | [Live](#) | TypeScript, Express, MongoDB, Redis (BullMQ), Docker

Jan 2026 – May 2026

- Automated README generation on every git push via GitHub Webhooks; **50 users, 68 active repositories, 959 automation runs at 85% success rate**; live at daemondoc.online with subscription billing.
- Architected a Redis (BullMQ) job pipeline with exponential backoff and 3-attempt fault tolerance; reduced per-repository token cost by **70%** by filtering irrelevant source files before prompt construction.
- Orchestrated **4 services** with Docker Compose; configured Nginx reverse proxy and GitHub Actions pipeline enabling zero-downtime deployments.

IIT Jodhpur PHC System | [GitHub](#) | Node.js, Express, Postgres, Prisma, Docker

Jan 2026 – Apr 2026

- Spearheaded backend architecture in a 3-member team; replaced paper-based workflow with a permission-based platform: **6 roles, 63 endpoints, 20 database models** with JSON Web Token and LDAP dual authentication across 7 Scrum sprints.
- Verified reliability via **48 test cases**, a 59-check smoke suite, and load tests at **20 concurrent users**; approved by professor for production deployment projecting **5 staff-hours saved per day**.

COURSEWORK

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|----------------------------------|-------------------------------|------------------------------|
| • Data Structures and Algorithms | • Database Management Systems | • Discrete Mathematics |
| • Object-Oriented Programming | • Software Engineering | • Pattern Recognition |
| • Operating Systems | • Machine Learning | • Probability and Statistics |

ACHIEVEMENTS

- JEE Advanced 2024: AIR **3449**; JEE Main 2024: **99.65** percentile, AIR **5680**.
- Completed **Hacktoberfest 2025** with 6 merged pull requests across open-source repositories including Wagtail.
- Awarded **Bharti Airtel Scholarship 2024** providing academic funding for 4 years at IIT Jodhpur.

LEADERSHIP

- Junior Core Team Member** at DevUp Labs, IIT Jodhpur; mentored 30 students and facilitated 2 open-source workshops.
- Core Member** at E-Cell, IIT Jodhpur; supported 3 entrepreneurship events and initiatives over 9 months.
- Core Member** at Programming Club, IIT Jodhpur; co-organized 5 contests and 10 sessions on competitive programming.